

SCOPE : L4 — Boundary

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Chapter 0: Introduction

The Sephirotic names used in this paper are not intended in a religious or mystical sense. Within SCOPE, they are borrowed as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for inferring the cognitive structures of humans and AI. Here, a coordinate system does not refer to a flowchart indicating that cognitive processing proceeds in a fixed order. Rather, it is a framework for organizing and inferring cognitive structure. Each layer does not indicate a hierarchical relationship or a temporal sequence.

Within SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is the output itself; it is not possible to directly know what cognitive structure was at work internally. For this reason, SCOPE uses observable output as a clue to infer the cognitive structure behind it.

In reality, there are cases in which a person is unable to refuse even when refusal would be appropriate. At the same time, there are cases of rejecting others more than necessary, or of being excessively suspicious. There are also cases in which a person can view themselves objectively, and conversely, cases in which it is difficult to maintain distance from oneself. These may not be fully explained by personality or by notions of good and bad alone.

This paper addresses L4, one of these cognitive structures. L4 is a cognitive layer that sets boundaries with objects of cognition. Here, objects of cognition include people, information, values, and the self. The definition and characteristics of L4 are discussed in detail in Chapter 1.

L4 is not defined as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any particular layer; rather, it treats cognition as a structure in which each layer influences the others.

This paper examines the characteristics and basic properties of L4, centered on its role as a boundary-setting layer, as well as the functional correspondence between humans and AI. Note that some points remain at the hypothesis stage and are left as topics for future consideration.

Chapter 1: What Is L4

L4 is a cognitive layer that sets boundaries with objects of cognition.

Here, a boundary refers to the function of verifying whether an object of cognition is "really true," and, on that basis, judging whether to let the object through or to hold it apart. Within

SCOPE, L4 is positioned as the cognitive layer that organizes and infers this function. A boundary with the object of cognition is formed as a result of this judgment.

Here, objects of cognition include people, information, values, and the self.

L4, as used here, does not refer to a biological organ or a specific region of the brain. Within SCOPE, it is positioned as a cognitive layer for inferring how the boundary between a person and an object of cognition is organized.

This paper positions L4 as the boundary-setting layer, and its characteristics are organized in the chapters that follow.

Chapter 2: Basic Characteristics of L4

2-1: L4 Judges Objects of Cognition

Within L4, an object of cognition is verified against the question "Is this really true?," and a judgment is made as to whether to let the object through or to hold it apart.

Here, judgment includes determining, within the person's own internal frame, what is right or wrong, good or bad, and superior or inferior. However, this does not guarantee that the judgment corresponds to reality or to objective fact. Even when something is judged internally to be "absolutely right," "clearly wrong," "good," "bad," "superior," or "inferior," that judgment may diverge from reality. How the reasoning or theory underlying such a judgment is constructed involves other layers and requires separate consideration; it is not addressed in this paper.

For example, upon encountering new information, a person may simply accept it, or may instead ask, "Is this really true?" and seek additional confirmation. Similarly, when faced with another person's proposal or with one's own thoughts, a person may pause to examine them rather than accepting them outright. These can all be considered examples of judgment carried out by L4.

Within SCOPE, this process — verifying whether an object of cognition is "really true" and judging whether to let it through or hold it apart — is positioned as the basic function of L4.

2-2: L4 Judges Based on Available Information

The judgment carried out by L4 is not always based on sufficient information. L4 verifies whether an object of cognition is "really true" and makes its judgment based on whatever information is available to the person at that time.

Here, available information includes events the person has experienced, knowledge acquired through learning, and information currently accessible to them. For this reason, even when the same object of cognition is being handled, the outcome of the judgment may differ depending on what information is available.

For example, when examining the same theory, the material available as grounds for judgment differs depending on whether multiple sources and counterarguments can be consulted, or only limited information is available. A judgment may also change once new information is obtained.

Within SCOPE, differences in the amount of information available, and in the conditions under which it is obtained, are positioned as one of the factors that influence judgment by L4.

2-3: L4 Forms Boundaries

The result of L4's judgment appears as a boundary with the object of cognition. A boundary is the form that the judgment — whether to let the object through or to hold it apart — takes as a relationship within the cognitive structure.

This boundary does not mean only rejection or exclusion. For example, keeping distance from another person, viewing oneself objectively, cross-checking information, or deliberately holding someone at a distance for the sake of their future can all be understood as results of setting a boundary with an object of cognition.

At first glance, rejection, objective self-observation, cross-checking, keeping distance, and holding someone at a distance appear to be different cognitive processes. Within SCOPE, however, all of these are positioned as phenomena arising from L4's judgment of whether to let an object of cognition through or to hold it apart.

For this reason, L4 is not a "layer of rejection." The role of L4 is to set boundaries, and rejection or exclusion is merely one of the phenomena observed as a result.

Chapter 3: L4 Changes

The judgment carried out by L4 is not always constant. Even with the same object of cognition, the judgment of whether to let it through or hold it apart may change depending on the relationship with the object, experience, or changes in circumstance.

For example, a person who was approached with caution and kept at a distance on first meeting may later be judged differently once a relationship of trust has formed. Conversely, someone who was once trusted may come to be kept at a distance as a result of new experience.

Likewise, even when the object of cognition itself does not change, acquiring new knowledge or experience increases the information available to the person, and a different judgment may result. As a consequence, the substance of verifying "is this really true?" may change, and the boundary may be revised.

This change, however, does not necessarily arise from L4 alone. The judgment may also change as part of a broader change across the cognitive structure; this interaction is addressed in the next chapter.

Within SCOPE, L4 is thus positioned not as a fixed property, but as a cognitive layer that can change according to the object of cognition, experience, and circumstance.

Chapter 4: L4 Interacts with Other Layers

The judgment carried out by L4 is not necessarily made in isolation. When verifying whether an object of cognition is "really true," cognitive information processed by other layers may be used as material for the judgment.

For this reason, even when L4 is dealing with the same object of cognition, the judgment of whether to let it through or hold it apart may change if the state of other layers differs. This does not mean that the function of L4 itself has changed; rather, it can be understood as a result of a change in the cognitive information used for the judgment.

For example, in response to the same remark, one person may let it through without objection, while another may keep their distance. Likewise, even the same person may judge differently as their experience or circumstances change. Such differences need to be understood not as L4 acting alone, but as the interaction of the cognitive structure as a whole.

Within SCOPE, each layer is defined as an independent cognitive function, while also being treated as part of a cognitive structure in which the layers influence one another. L4 is likewise positioned as a cognitive layer that functions within this interaction with other layers.

As the purpose of this paper is to organize the role of L4 on its own, the detailed processes by which each layer influences this judgment are not addressed here. These are intended to be organized separately in a paper covering SCOPE as a whole.

Chapter 5: L4 and Its Relationship to Malkuth

Within SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as Malkuth. For this reason, L4 itself cannot be directly observed; what can be observed is the output that appears as the result of L4's judgment.

For example, when behaviors such as keeping distance from an object of cognition or viewing oneself objectively are observed, these can serve as clues suggesting that L4 has been at work. However, the state of L4 cannot be uniquely determined from these phenomena alone, since the same output may also arise from a different cognitive structure.

For this reason, SCOPE does not determine the state of L4 from observed output alone. Using the information observed at Malkuth as a clue, the state of L4 is inferred by cross-referencing it with other cognitive information.

In this way, L4 is positioned not as a cognitive layer directly observed at Malkuth, but as a cognitive layer inferred on the basis of observable output.

Chapter 6: Correspondence with LLMs

The function addressed by L4 — verifying an object of cognition and judging whether to let it through or to hold it apart — is not limited to humans alone. In LLMs as well, behavior is observed in which input information is not simply adopted as given, but is checked before a response is generated.

For example, behaviors such as checking input information for contradictions, or not adopting the input as given, can be seen as functionally corresponding to the workings of L4 in humans. Moreover, the object is not limited to information input from outside. Behavior is observed in which, among multiple possible candidate responses, a single response is produced as output. This suggests that a similar judgment of whether to let content through may extend to content the system itself has generated.

This does not mean, however, that L4 in humans and the internal processing of LLMs are identical. What this paper addresses is the possibility of correspondence as a cognitive function, not a claim of identity at the level of biological or computational implementation.

Within SCOPE, L4 is positioned as a cognitive coordinate for comparing and organizing the cognitive structures of humans and LLMs on the basis of this functional correspondence.

Conclusion

This paper has organized L4 as a cognitive layer that verifies whether an object of cognition is "really true" and judges whether to let it through or to hold it apart.

L4 is not a cognitive layer whose purpose is rejection or exclusion. By setting boundaries with objects of cognition, it was shown that L4 can give rise to a variety of phenomena, including keeping distance and viewing oneself objectively. It was also shown that this judgment is not fixed, and can change according to the object of cognition, experience, circumstance, and interaction with other layers.

Furthermore, the possibility of a functional correspondence with the workings of L4 was examined on the basis that, not only in humans but also in LLMs, behavior is observed in which an object of cognition is not simply adopted as given, but is output only after undergoing verification and judgment.

L4 is a layer that judges whether to let an object of cognition through or to hold it apart. However, how an object of cognition that has been judged comes to be accepted, and how it connects to subsequent cognitive processing, cannot be explained by L4 alone.

The next paper will focus on L5 to organize how an object of cognition that has passed through L4's judgment comes to be accepted, and how it connects to subsequent cognitive processing.